TS2000
TS2100
Robot Controller

ETHERNET INTERFACE MANUAL

Rev.3
April 1, 2020

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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The information contained in this manual is subject to change without notice to effect improvements.

Preface

This manual describes the functions and handling method when the TS2000 robot controller is connected with an external equipment via Ethernet interface.
The Ethernet interface is the optional function adding a diversity of network functions to the robot controller.

* Instruction manuals which are referred to from this manual

- Communication Manual
- TSPC Manual
- Robot Language Manual
- Operation Manual
- Maintenance Manual

 **CAUTION**	This manual does not contain any detailed descriptions on power and robot connection. For the connection of the power and robot, see the Installation & Transport Manual.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damages to assets and to assure correct use.
Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries 1) or physical damage 2)."

1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.

 **CAUTION**	To perform the work ranging from robot installation to operation with safety, read through and through the Safety Manual provided separately before actually starting the work.

Maintenance and Inspection

To use the robot safety, strictly observe the following matters.

! DANGER	
 Prohibited	• NEVER burn, disassemble or charge the battery. Otherwise, it may explode.
 Mandatory	• Before performing the maintenance and inspection, be sure to turn off the main power switch of the controller. • When disposing of batteries, be sure to follow the user's regulations.

! CAUTION	
 Disassembly prohibited	• The user should NEVER replace or change parts other than those stipulated in the instruction manual. Otherwise, the performance will deteriorate, resulting in troubles.
 Mandatory	• To replace parts, use the spare parts designated by SHIBAURA MACHINE. • Carry out the maintenance and inspection on a regular basis. Otherwise, the equipment may go wrong or accidents will be caused.

! CAUTION	
	To perform the maintenance and inspection of the robot with safety, read through and through the Maintenance Manual provided separately before actually starting the work.

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1. Preparations

This section describes the preparations for starting the Ethernet function. This function is optional. When it is used, setting of various parameters and X8HL option printed board are necessary.
The X8HL printed board is an optional printed board for the system robot controller (TS2000/TS2100). It is incorporated in the option connector of the main controller. When the optional printed board is already built in the controller, skip the assembling work.

1.1 Printed Board Assembling Procedures

Assemble the X8HL printed board in the robot controller according to the following procedures.

a) Turn off the power of the robot controller.

b) Turn off the main power (200 V) supplied to the robot controller. To assure safety, disconnect the power connector (ACIN CN1).

c) Remove the top cover of the robot controller.
This cover is secured with a total of nine (9) screws (i.e., four (4) pcs. each on the sides of the robot controller, and one (1) pc. on the top). Remove these screws, using a screwdriver, then draw the cover toward you and dismantle.

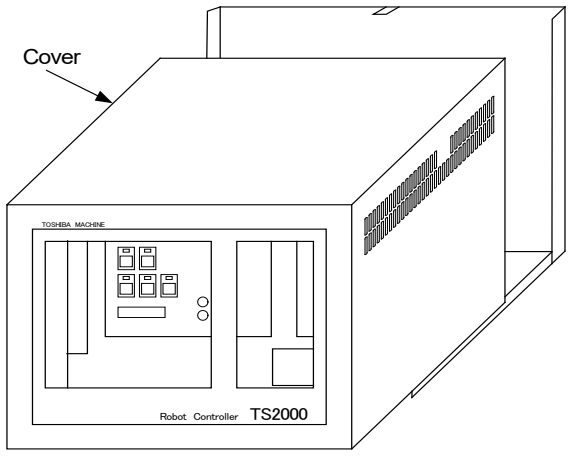


Fig. 1 Removing the cover

d) Remove the dummy connector plate attached to the rear side of the robot controller.

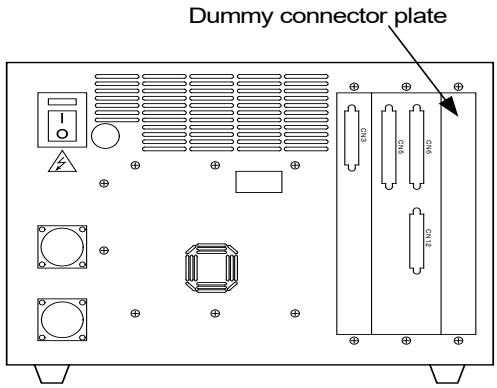


Fig. 2 Rear side of robot controller

e) Insert the X8HL printed board to the mother board.
Remove the connector cover of the mother board (X8HB) connector CN6, then completely insert the X8HL printed board to CN6 of the X8HB printed board. After the insertion, secure the rear side (two (2) places) and internal steel plate (one (1) place) of the robot controller with M3 screws.

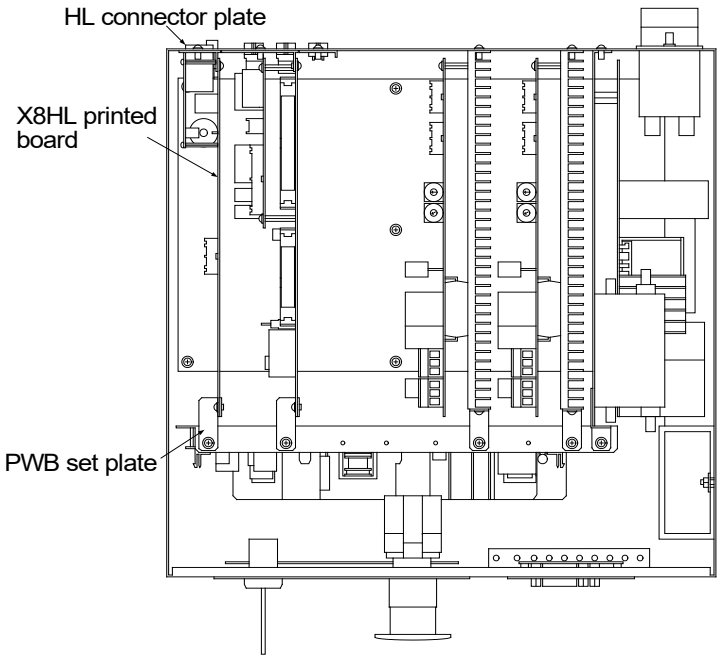


Fig. 3 Layout of robot controller panel interior

f) Mount the cover on the robot controller as originally set, and secure with the screws.

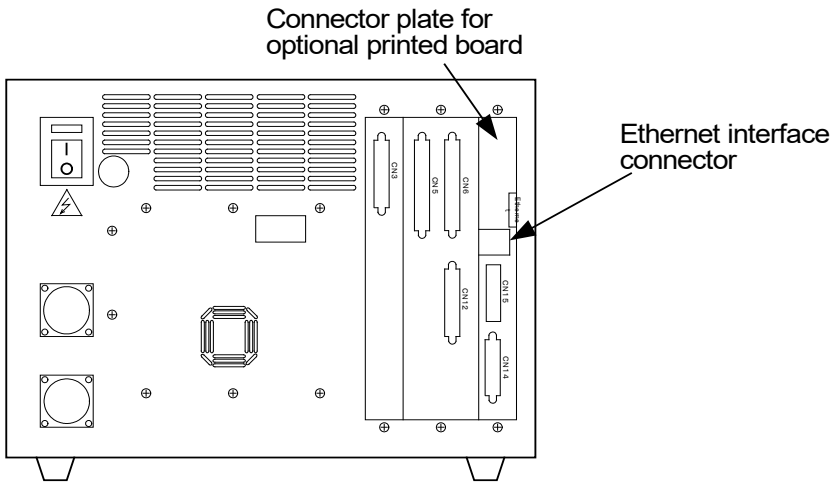


Fig. 4 Rear view of robot controller after assembly

1.2 Connecting Ethernet Cable

When the hub is used, a straight cable is used. When directly connecting a single personal computer with a single controller, a cross cable is used.
Connect the cable with the "Ethernet" connector provided at the rear side of the controller as assembled in Para. 1.1 above.

2. Ethernet Interface

2.1 Ethernet Interface Functions

(1) Connection is supported pursuant to the "10base–T".

(2) Communication with a personal computer and peripheral equipment on the Ethernet is possible in the same local area, using the "TCP/IP" protocol.

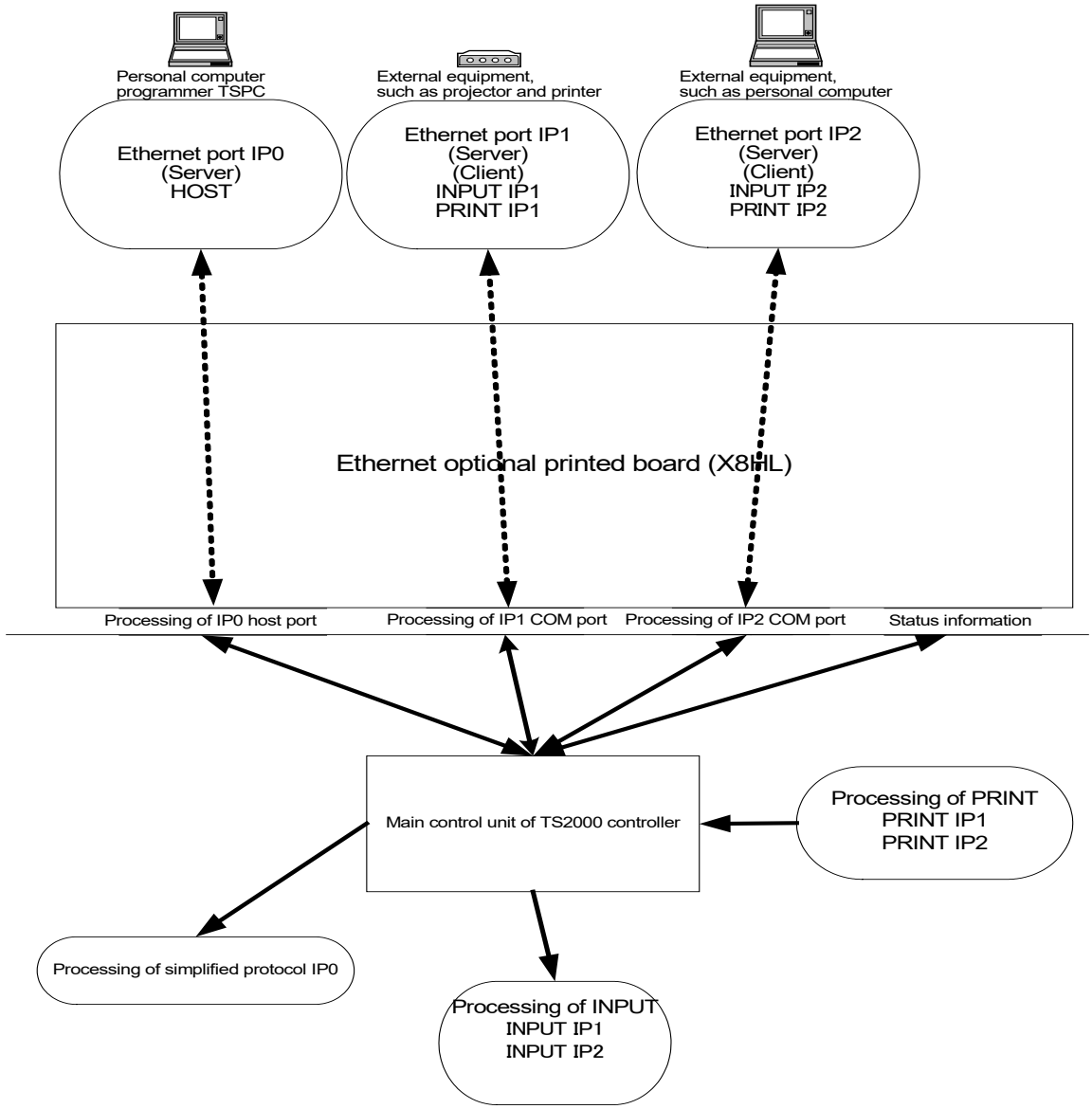
(3) When the Ethernet is used, the following functions can be used.

[1] Program creation and editing, and program maintenance by starting up the debug program are possible, using personal computer programmer TSPC.

[2] Numeric and position data can be transferred, using the robot program. Data input and output are possible, using the INPUT and PRINT statements of the SCOL language.

Example:

```
INPUT   IP1, INDATA
PRINT   IP1, OUTDATA, CR
```


Personal computer
programmer TSPC
External equipment,
such as projector and printer
External equipment,
such as personal computer
Ethernet port IP0
(Server)
HOST
Ethernet port IP1
(Server)
(Client)
INPUT IP1
PRINT IP1
Ethernet port IP2
(Server)
(Client)
INPUT IP2
PRINT IP2
Ethernet optional printed board (X8HL)
Processing of IP0 host port
Processing of IP1 COM port
Processing of IP2 COM port
Status information
Main control unit of TS2000 controller
Processing of PRINT
PRINT IP1
PRINT IP2
Processing of simplified protocol IP0
Processing of INPUT
INPUT IP1
INPUT IP2

3. Parameter Setting

Before using the Ethernet interface, the option parameter described in the MACHINE.PAR file and Ethernet parameters described in the ETHERNET.PAR file should be specified.

3.1 Setting Machine Parameter

Perform the ON/OFF setting for the option switch as registered in the MACHINE.PAR file.

No. 5 1 ····· COM2 ON
2 ····· Ethernet ON
3 ····· COM2 and Ethernet ON

Example: [M03] Option switch
= 0 0 0 0 3 0
= 0 0 0 0 0 0

3.2 Setting Ethernet Parameters

3.2.1 [E00] Open mode IP0 IP1 IP2 IP3
{0: non 1: Robot is TCP server 2: Robot is TCP client}

Perform the ON/OFF setting for the Ethernet function.

0 : Unused (not open)
1 : Serves as the TCP server.
2 : Serves as the TCP client.

As the IP0 is exclusively used for the simplified protocol, however, designate the server mode. If the external equipment is used as the server, specify "client" for the robot controller. If it serves as the client, specify "server" for the controller.

Example:
= 1 0 0 0

3.2.2 [E01] Robot controller IP address

Specify the robot controller's own IP address.
The IP address is an address for identifying the controller, and consists of four (4) numeric numbers from 0 ~ 255 and periods flanked by the numbers. When directly connecting a single controller with a single personal computer connected to the network, the IP address can be the initial value (any value). When connecting them via LAN, however, specify the IP address according to the instructions given by the customer's LAN system manager. If the same IP address is used repeatedly, the Ethernet function will not work properly. The personal computer communicating with the robot controller should be connected on the same network.

Example:
= 192.168.0.124

3.2.3 [E02] Robot controller name (To be used when the function is extended)

Specify the name of the robot controller.
This parameter is slated for use when the Ethernet function is extended.

Example:
= "N01–TS2K"

3.2.4 [E03] Subnet mask

Specify the subnet mask of the network to be connected. Normally, the initial value can be specified. When connecting via LAN, specify the subnet mask according to the instructions given by the customer's LAN system manager.

Example:
= 255.255.255.0

3.2.5 [E05] Default gateway (To be used when the function is extended)

Specify the default gateway address.
This parameter is slated for use when the Ethernet function is extended.

Example:
= 192.168.0.1

3.2.6 [E06] Own port no

Specify the port number of each port used by the robot controller. The port number is something like the frequency of a radio. Specify the port number for each of IP0 ~ IP3. Normally, the initial value need not be changed. Take careful precautions not to use the same port number repeatedly.

Example:
{IP0}
= 1000
{IP1}
= 1001
{IP2}
= 1002
{IP3}
= 1003

3.2.7 [E07] Port number of destination

Specify the port number of each port used by the destination. This mode is valid only in the client mode. These port numbers should be made identical with the port numbers used by the destination server.

Example:
{IP0}
= 1000
{IP1}
= 1001
{IP2}
= 1002
{IP3}
= 1003

3.2.8 [E08] IP address of destination

Specify the IP address of each port connecting the destination. This mode is valid only in the client mode.

Example:
{IP0}
= 192.168.0.150
{IP1}
= 192.168.0.151
{IP2}
= 192.168.0.152
{IP3}
= 192.168.0.153

3.2.9 [E09] Robot community name (To be used when the function is extended)

Specify the community name of the network.
This parameter is slated for use when the Ethernet function is extended.

Example:
= "N01–TS2K"

3.2.10 [E10] DHCP (To be used when the function is extended)

Specify whether the address should be acquired from the DHCP server.
This parameter is slated for use when the Ethernet function is extended.

Example:
= 0

3.2.11 [E11] Host mode FULL control port

Specify the RS232C or TCP/IP from which the robot run can be controlled while the controller is in the host mode.
{0: RS232C 1: Ethernet}

Example:
= 0

* Example of parameter setting

```
COPYRIGHT(C) 2003 BY SHIBAURA MACHINE CO.,LTD.
ALL RIGHTS RESERVED.
OPTION TCP/IP INTERFACE PARAMETER FILE "ETHERNET.PAR" FOR TS–2000
[E00] OPEN MODE IP0 IP1 IP2 IP3
{0:NON 1:ROBOT IS TCP SERVER 2:ROBOT IS CLIENT}
= 1    0    0    0
[E01] ROBOT CONTROLLER IP ADDRESS
= 192.168.0.124
[E02] ROBOT CONTROLLER NAME
= "NO1-TS2K"
[E03] SUBNET MASK
= 255.255.255.0
[E04] DEFAULT GATEWAY
= 192.168.0.1
[E05] OWN PORT NO
{IP0}
= 1000
{IP1}
= 1001
{IP2}
= 1002
{IP3}
= 1003
[E06] PORT NUMBER OF DESTINATION
{IP0}
= 1000
{IP1}
= 1001
{IP2}
= 1002
{IP3}
= 1003
[E07] IP ADDRESS OF DESTINATION
{IP0}
= 192.168.0.150
{IP1}
= 192.168.0.151
{IP2}
= 192.168.0.152
{IP3}
= 192.168.0.153
[E08] ROBOT COMMUNITY NAME
= "NO1-TS2K"
[E09] DHCP
= 0
[E10] HOST MODE FULL CONTROL PORT
{0:RS232C 1:ETHERNET}
= 0
[END]
```

4. Confirming the Connection

4.1 Confirming the Preparations

Before using the Ethernet function, confirm the following matters.

(1) Make sure that the Ethernet cable between the robot controller and external equipment (personal computer, for instance) is connected properly.

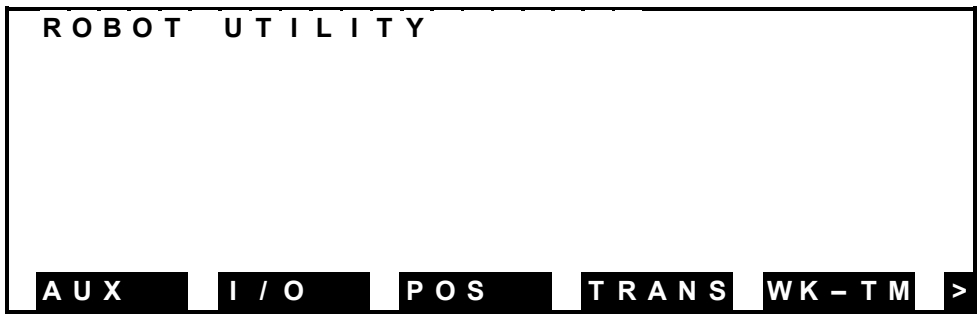
(2) Make sure that the Ethernet cable used is correct.
When directly connecting the external equipment and robot controller, a cross cable should be used. When the hub is used, a straight cable should be used.

(3) Make sure that the parameters of the controller are set legally.

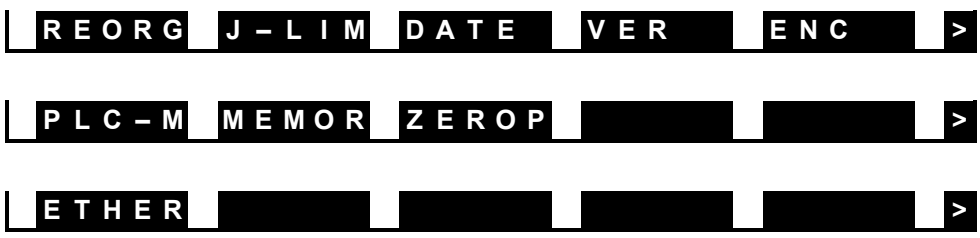
(4) Make sure that the controller power was turned off once after the parameter setting.

4.2 Confirmation through Ethernet Utility Display

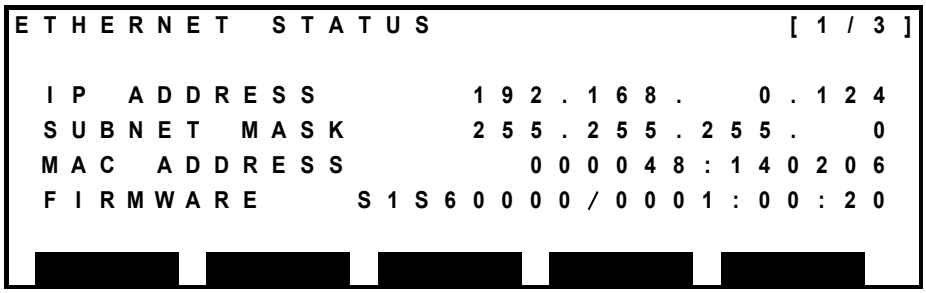
Press the UTILITY key, and the system calls the following display



Press the NEXT key three (3) times, and [ETHER] menu appears.

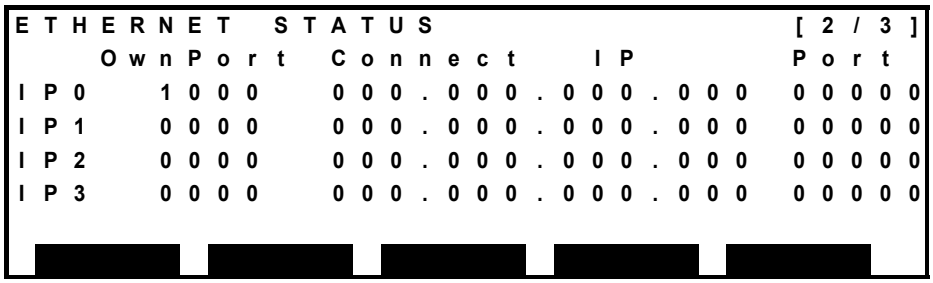


Press the [ETHER] key, and the system calls the following display.



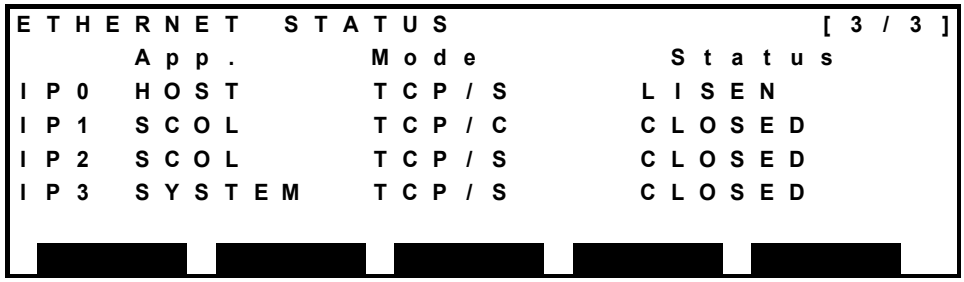
IP ADDRESS : Controller's own Ethernet address.
SUBNET MASK : Subnet mask.
MAC ADDRESS : MAC address.
FIRMWARE : Ethernet unit firmware version.

On this display, make sure that the controller's own Ethernet address and subnet mask as specified by the parameters beforehand are set.



IP0 ~ IP3 : Ethernet channel number.
OwnPort : Port number on the robot side.
Port : Port number of the destination.
Connect IP : IP address of the destination.

On this display, status information on connection when the controller is connected with external equipment is shown.



IP0 ~ IP3 : Ethernet channel number.
App. : Application name using each channel.
Mode : Type of protocol, open status.
Status : Change in Ethernet port status.

On this display, the operating status of each channel is shown.
Under the column of "App", the application name using each channel is displayed.
"HOST" signifies the connection pursuant to the simplified protocol.
"SCOL" represents the connection when the robot is controlled, using the robot language and pursuant to the non-procedural protocol.
"TCP/S" under the column of "Mode" signifies the server connection as per the TCP protocol.
"TCP/CS" under the column of "Mode" means the client connection as per the TCP protocol.
Under the column of "Status", change in Ethernet status is displayed. For details on each status, refer to the TCP/IP manual, etc., commercially available.

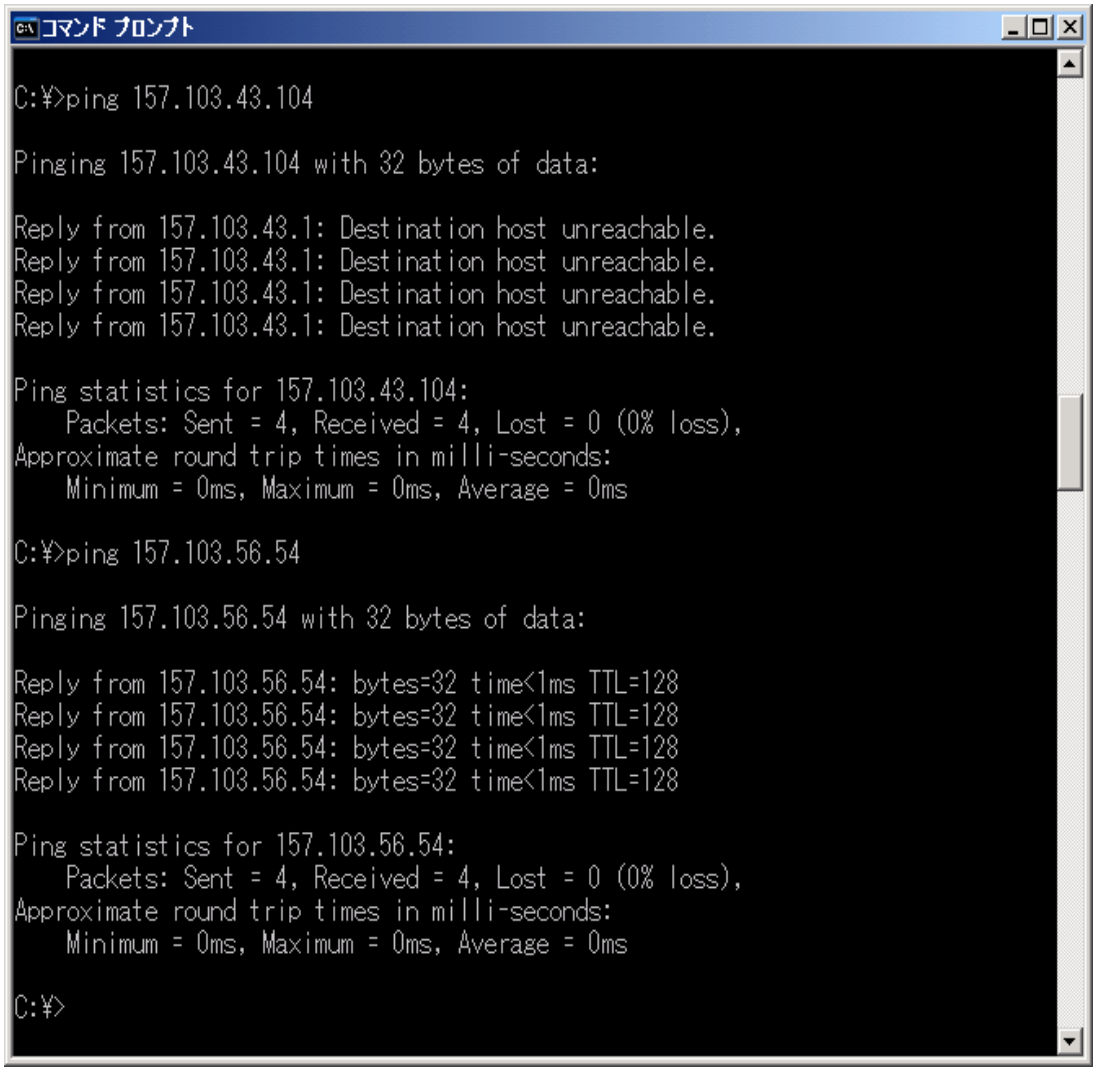
1 : Closed
2 : Listen
3 : Syn-Sent
4 : Syn-Received
5 : Established
6 : Fin-Wait1
7 : Fin-Wait2
8 : Close wait
9 : Closing
10 : Last ACK
11 : Time Wait

4.3 Confirmation through Windows Personal Computer

This paragraph explains the confirming method of connection, using the "ping" command of the Windows.
Start "Command prompt" from the "Start" menu of the Windows, then specify the IP address of the robot controller in the manner given below.

C:¥> ping IP address of robot controller

If an error has occurred during communication, "unreachable" or "Request Time out" is displayed as shown on the upper side of the screen below.
If the communication is normal, "Reply from ···" is displayed as shown on the lower side of the screen.



5. Send-Receive Protocols with TCP/IP Device

5.1 Non-Procedural Protocol

This protocol is used when the data is transferred with the external equipment, using the SCOL language. The external equipment is connected via socket communication. When transferring the data with the device on the network, the controller can serve as the TCP/IP client or server.
When the external equipment serves as the server, the controller should be set to "client". Likewise, when it is used as the client, the controller should be set to "server".
The data format of character strings transferred between the robot controller and TCP/IP device is as follows:

Character string + CR(13)	ASCII format. "13" (carriage return) of ASCII character is added to the character string data.
Character string, Character string + CR(13)	When transmitting multiple data, each character string is separated with a comma.

Data transfer with the robot language program is performed pursuant to the non-procedural protocol.
The robot language provides the two (2) commands of PRINT and INPUT for data communication. No check is performed, however, on the reply to the data received by the robot controller and time-up.

5.1.1 Communication with Robot Language

Data communication is performed with the robot language program, using the PRINT and INPUT commands of the robot language.
For details on the robot language, see the robot language manual provided separately.

5.1.2 PRINT Command

Any character strings can be transmitted from the robot controller by using the PRINT command. The controller sends the character strings and variables as designated by the PRINT command.
The programming format of the PRINT command is as follows:

PRINT [<Channel>,] {<Character string> | <Expression> |} [, {<Character string> | <Expression>}] ····· [, CR]

<Channel> : Specify the communication port through which data is to be sent. Specify either of the following channels.
IP1 : Ethernet IP1 port
IP2 : Ethernet IP2 port

<Character string> : Specify the desired character strings flanked by ["].

<Expression> : Specify the expression consisting of a constant, variable, or combination of these with arithmetic operator and function.

CR : Specify when the record end code (0DH) is added to the end of the transmitted data.

Example: PRINT IP1, "INPUT DATA =", –1000.0/3, CR

In the PRINT command, two (2) or more character strings and expressions separated by [,] can be specified. The content of the character strings as specified by codes ["] is transmitted in ASCII code. The result of operation of expression is transmitted in a fixed length of twelve (12) characters right-justified. If the value of the expression is an integer, it is transmitted by a decimal number consisting of up to ten (10) digits. If the value of the expression is a real number, it is transmitted by a maximum of seven (7) digits (i.e., a maximum of four (4) digits of integer and a maximum of three (3) decimal places) for a maximum total of eight (8) characters including a decimal point. A one (1) character of sign (i.e., plus or minus) is provided at the top of the number. The plus (+) sign can be omitted. The number consisting up to twelve (12) characters is right-justified and any unused character spaces is filled with a space code (20H).

All data are transmitted in ASCII code. Commas which were specified by separating multiple character strings and expressions are not sent. When a plural number of character strings and expressions are specified in the PRINT command, they are transmitted as one (1) text from the robot controller.

Example: When the above program has been executed, the following data is transmitted.
INPUT DATA = –333.333 (CR)

Note: (CR) signifies the record end code (0DH).

5.1.3 INPUT Command

While the PRINT command is used to transmit data, the INPUT command is used to receive the data. Only a real number or integer can be received by the robot controller. Data received by the controller is substituted to the variable designated by the INPUT command. The received data can be used by referring to this variable in the robot language program.

INPUT [<Channel>,], <Variable>, [<Variable>] ·····

<Channel> : Specify the communication port through which data is to be received. Specify either of the following channels.
IP1 : Ethernet IP1 port
IP2 : Ethernet IP2 port

<Character string> : Specify the desired character strings flanked by ["].

<Variable> : Specify the variable where received data is to be substituted.

Example: INPUT IP1, N1, N2

In the INPUT command, two (2) or more variables can be specified by separating them with commas. In the INPUT command, the controller waits until the data reaches from the specified communication channel. At the end of the data sent from the controller, the record end code (0DH) should be specified. When sending multiple data to the controller, such data can be separated with commas.

When the number of data received by the controller is larger than the number of data specified by the INPUT command, the surplus data is ignored and used at the next request for the INPUT command. When the number of data received by the controller is smaller to the contrary, the INPUT command waits until the remaining data reaches.

Note 1: The robot controller reads the data it has received after the start of the INPUT command execution.
The data is stored in the ring buffer, and the input data is read according to the requirement of the INPUT statement. As the data received before execution of the INPUT command may be neglected, take careful precautions on the timing of data transfer.

Note 2: If the program execution has stopped while the controller was executing the INPUT command and waiting for arrival of data, execution of the INPUT command is interrupted. The program is restarted from the step next to the INPUT command. Any data which is yet to reach at this point of time is processed as zero (0).
When creating a program using the robot language, there should not be any problem even if the data received by data communication has been processed as zero (0).
For this purpose, if the received data is zero (0), program so that the data request can be made again to the host computer, or program in such a manner that check of the received data can be done by adding a checksum data to the received data. (For the specific example program, see Para. 5.1.5.)

Note 3: When the INPUT command has been executed during the step operation in the test operation mode, the program execution is suspended until the data reaches. This is also applied when the INPUT command has been executed in the direct mode.

Note 4: When two (2) or more variables have been specified in the INPUT command, transmit all data as separated with commas in one (1) text from the host computer.

5.1.4 Clear of Communication Buffer

If the number of data received by the controller is larger than the number of data specified by the INPUT command, the surplus data is stored in the communication buffer, which is used when the INPUT command is requested at the next time. If any data is left in the communication buffer, unexpected data reaches when the INPUT command is executed. To clear the data left in the communication buffer, therefore, character string "BUFFRESET" should be output to the appropriate communication port.

Example: When clearing the communication buffer of the IP1 port,
PRINT IP1, "BUFFRESET"

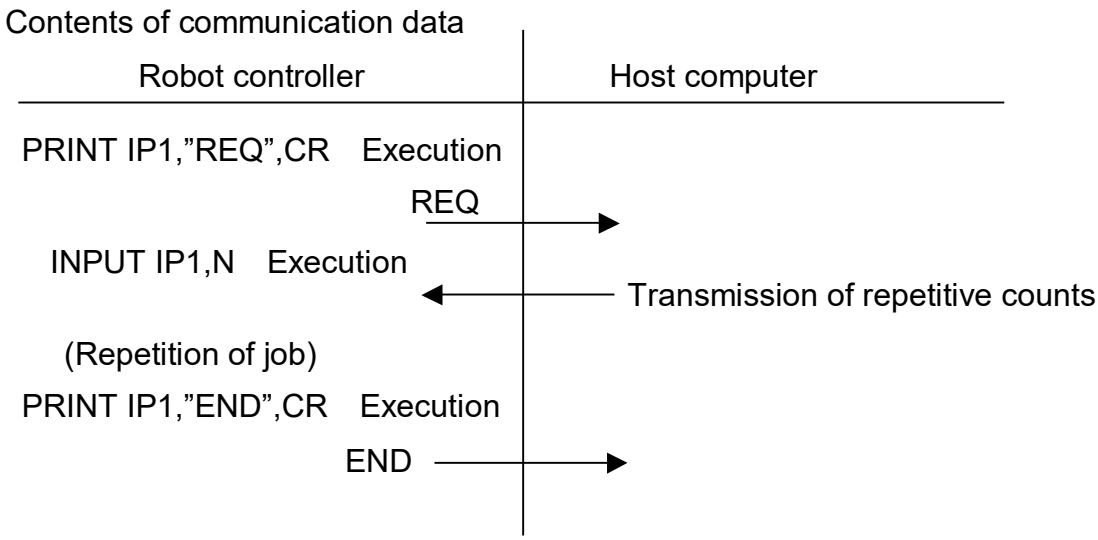
5.1.5 Use of Data Communication with Robot Language Program (Example)

A program can be branched out and the robot position can be compensated via data communication with the robot language program.
A usage is exemplified below.

* Example programs
  The number of repetitive counts of motion and the job to be executed can be selected by means of the INPUT and PRINT commands.

```
PROGRAM REPEAT
   N=0
START:
PRINT IP1,”REQ”,CR
INPUT IP1,N
IF N==0 THEN GOTO START
FOR K=1 TO N
MOVE A1
MOVE A2
....
NEXT K
PRINT IP1,”END”,CR
END
```

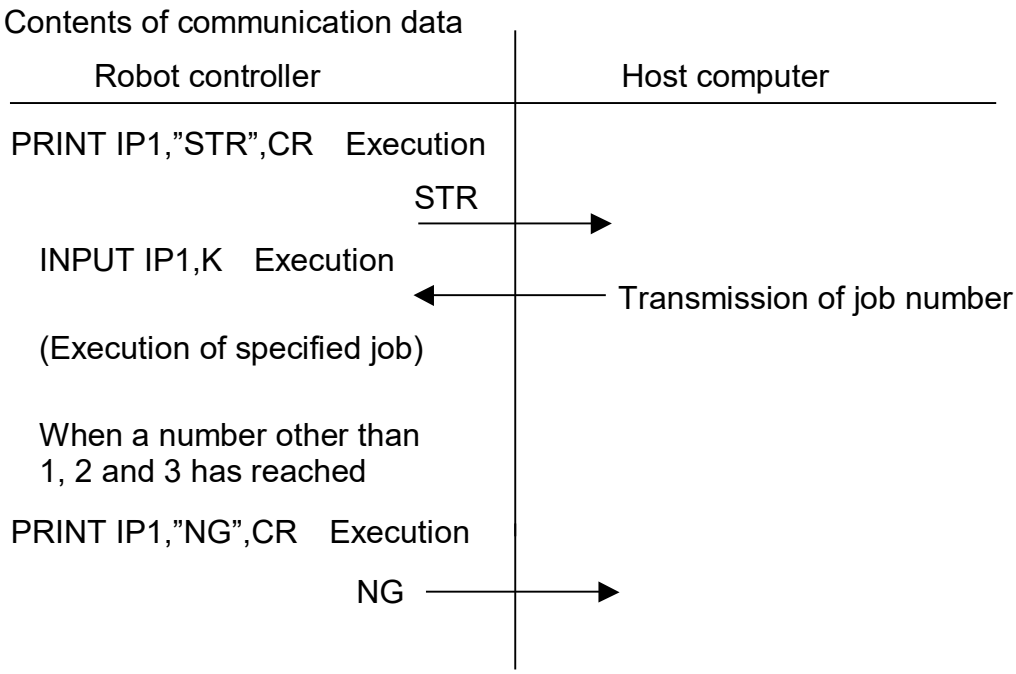
The job is repeated by the counts specified by the host computer.
After the host computer has received character string "REQ" from the robot controller, it sends the number of repetitive counts. The controller reads the number of repetitive counts to variable N and repeats the job as per the FOR statement.
If the program execution has been interrupted while the controller was waiting for the arrival of data in the INPUT command, variable N is processed as zero (0). To cope with this situation, program so that the controller branches to the head of the program to request re-sending of the data to the host computer.
At the end of the job, character string "END" is sent to the host computer.



```
PROGRAM SELECT
      K=0
START:
   PRINT IP1,”STR”,CR
SELECT:
   INPUT IP1,K
   GOTO(K) L1,L2,L3
   PRINT IP1,”NG”,CR
   GOTO FIN
L1:
   (Job 1)
   GOTO FIN
L2:
   (Job 2)
   GOTO FIN
L3:
   (Job 3)
   GOTO FIN
FIN:
   END
```

The job to be executed is specified by the host computer.

After the host computer has received character string "STR" from the robot controller, it sends the number for designating the job. The controller reads the value to variable K and branches the program according to this value, using the GOTO ( ) command.
When the number is "1", job 1 is executed.
Likewise, when the number is "2" and "3", job 2 and job 3 are executed, respectively.
When the number is other than 1, 2 and 3, character string "NG" is sent to the host computer.
If the program execution has been interrupted while the controller was waiting for the arrival of data in the INPUT command, variable N is processed as zero (0). In this occasion, too, the controller sends character string "NG" to the host computer.

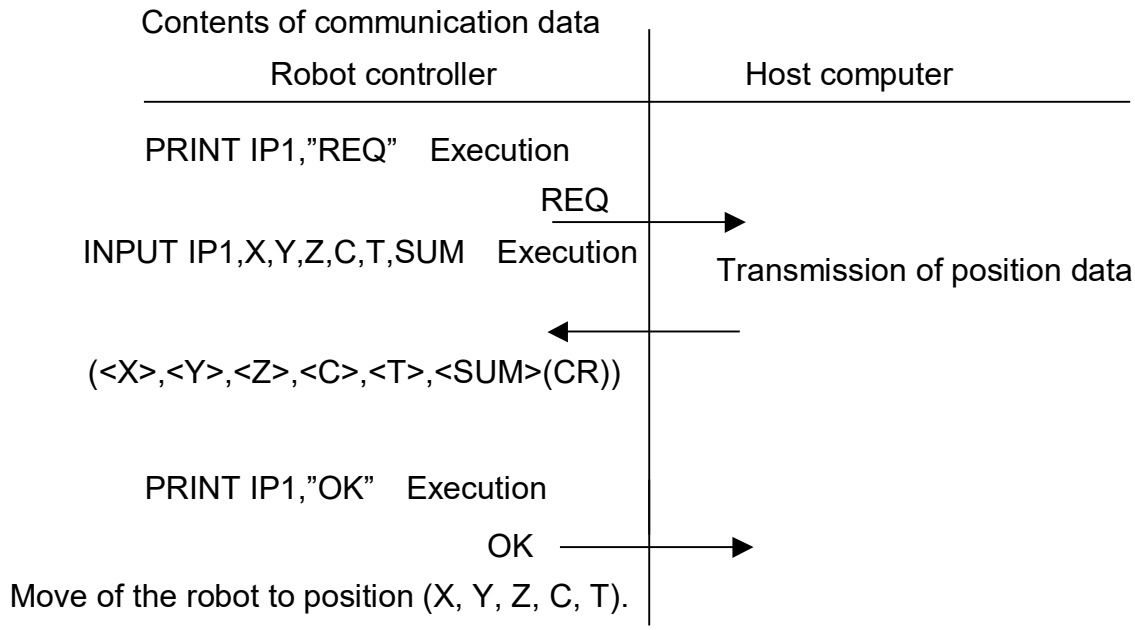


The data received from the host computer can be used to compensate the robot position.

[Direct designation of position]

```
PROGRAM DIRECT
   X=0.0
   Y=0.0
   Z=0.0
   C=0.0
   T=0.0
   PRINT IP1,”REQ”,CR
   INPUT IP1,X,Y,Z,C,T,SUM
   IF SUM==X+Y+Z+C+T+1
            THEN GOTO ACTION
   PRINT IP1,”NG”,CR
   GOTO FIN
ACTION:
   PRINT IP1,”OK”,CR
   P1=POINT(X,Y,Z,C,T)
   MOVE P1   WITH CONFIG=LEFTY
FIN:
   END
```

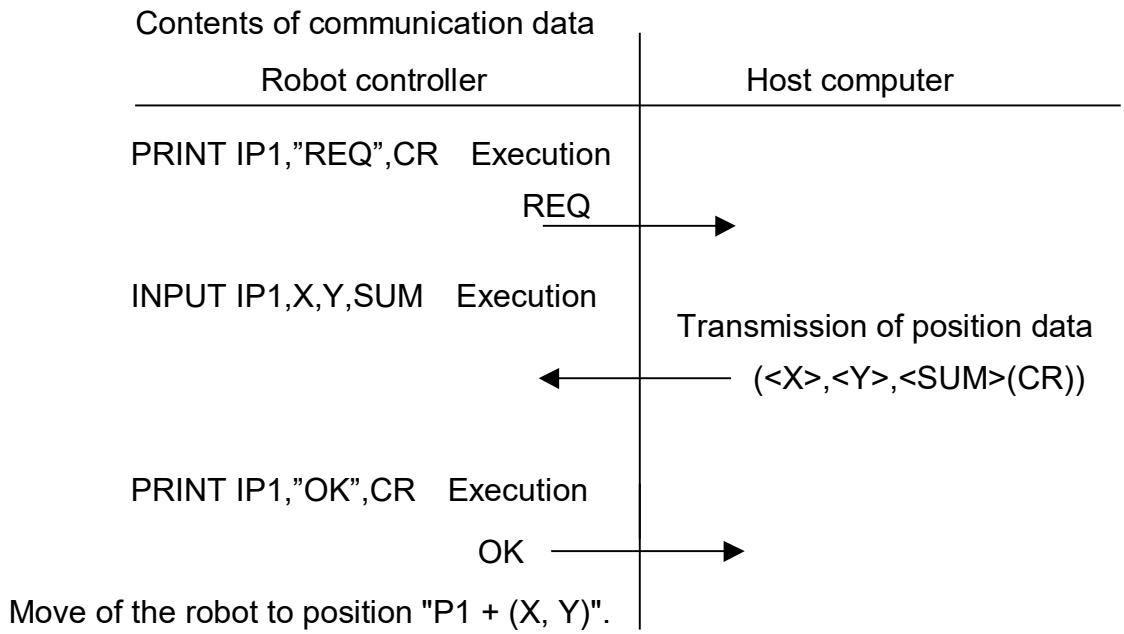
The robot moves to the position specified by the host computer.
After the host computer has received character string "REQ" from the robot controller, it transmits the position data X, Y, Z, C, T and the value which is the sum of these values added by "1" as the checksum data.
The controller compares the sum of the received data added by "1" with the checksum data. If the result is OK, the controller sends character string "OK". If an error is found in the received data, the controller sends character string "NG" and ends the program.
The controller creates position data P1 from the received data and moves the robot there.



[Designation of relative position]

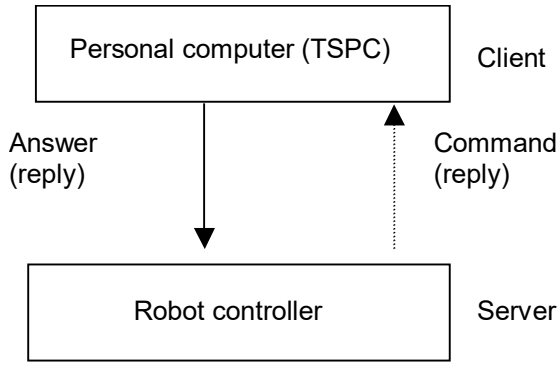
```
PROGRAM RELATIV
    X=0.0
    Y=0.0
    SUM=0.0
    MOVE P1
    PRINT IP1,”REQ”,CR
    INPUT IP1,X,Y,SUM
    IF SUM==X+Y+1
            THEN GOTO ACTION
    PRINT IP1,”NG”,CR
    GOTO FIN
    ACTION:
    PRINT IP1,”OK”,CR
    MOVE P1+POINT(X,Y)
FIN:
END
```

The robot moves to the position specified by the host computer.
After the host computer has received character string "REQ" from the robot controller, it transmits the coordinates X, Y and the value which is the sum of these values added by "1" as the checksum data.
The controller compares the sum of the received data added by "1" with the checksum data. If the result is OK, the controller sends character string "OK". If an error is found in the received data, the controller sends character string "NG" and ends the program.
The controller moves the robot to the position which is "coordinates X, Y of position data P1 + coordinates X, Y as received from the host computer."



5.2 Simplified Procedural Protocol

IP0 can be located at a mid-position between the host computer and robot controller by parameter setting. After the power is turned on, it starts as the TCP server at the port designated by the parameter which is set in the robot controller beforehand, and is waiting for a request sent from the personal computer (TSPC).
In principle, the robot controller serves as the client to the personal computer, and sends back a required data when connected with the personal computer.
When the connection command for communication has reached from the TSPC, the connection starts. Likewise, when the disconnection command has reached from the TSPC, the connection ends.



5.2.1 Setting of TSPC

The version of TSPC designed for the Ethernet interface is 1.06 or over.
Select [Option] menu → [Set com.] menu.
For the connecting method, select [TCP/IP].
For the setting of TCP/IP connection, specify the IP address of the destination controller and the port number as set in ([E06] Own port no {IP0}) beforehand.
For the operation, see the TSPC manual provided separately.

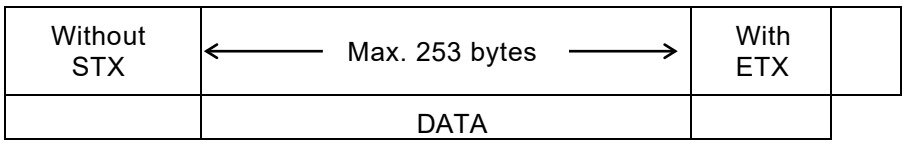
5.2.2 Transmission Format

In this communication, data are transmitted text by text, as shown below. Maximum real data size that can be transmitted in one (1) text is 253 bytes. Real data exceeding this size should be transmitted in the following manner.

(1) File (robot program, position data, parameter)
A file should be transmitted after divided into a few texts. In reply to each text transmission, an answer is sent from the receiving station. For the 2nd and after text transmission, no file upload/download command is required.

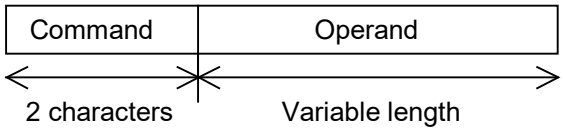
(2) Message
A message should be transmitted through a few counts of communication. That is, the data should be divided into a few texts and each text should be sent as an independent data. The data receiving station reconstructs these text data into one (1) message.

- Text per packet (No STX is included. ETX is added at the last.)



Text length: Max. 254 bytes
Content of text:
DATA Data + ETX (max. 254 bytes)

- Data format
The basic text format is as follows:



(1) Command
Specify the type of the command by two (2) alphabets.
For the type and content of each command, see the communication manual provided separately.

(2) Operand
The operand can vary with the type of the command. For details, see the communication manual provided separately.

(3) Characters
Use the alphanumeric characters and symbols of ASCII code.

- Alphanumeric characters
a b c d e f g h i j k l m n o p q r s t u v w x y z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
0 1 2 3 4 5 6 7 8 9

- Symbols
: " ' ( ) + – * / , . < > = ! [ ] { } % ^ & ? ;

5.2.3 Commands

The commands that can be used in this communication are shown in Table 1 and Table 2.

Table 1 Command list (Host computer → Robot controller)

No.	Command	Description	Details	Text
1	RN (*3)	Start	Automatic operation	RN
2	SP	Stop		SP
3	BR	Servo OFF		BR
4	SO	Servo ON		SO
5	RS	Reset	Program reset	RS, PRG
			Step reset	RS, STP
			Cycle reset	RS, CYC
			Output signal reset	RS, SIG
			Select reset	RS, SEL
6	SL	Program select		SL, File name
7	UL	File name	RAM file upload	UL, File name
8	DL	File name	RAM file download	DL, File name
9	CA	Directory	Directory request	CA
10	SU	Status	Status request	SU
11	EU	Error information	Error history upload	EU
12	ER	File erase		ER, File name
13	FL	File		FL, File content
14	OK	Acknowledge		OK
15	NG	Non-acknowledge		NG
16	EC (*2)	Internal command	Internal command transmit	EC, Internal command
17	JG	Jog	Job motion	JG, Axis name and direction

No.	Command	Description	Details	Text
18	MD	Guide mode setting	Guide mode setting	MD, Guide mode
19	RT	Guide speed (rate) setting	Guide speed (rate) setting	RT, Guide speed
20	SC	Guidance coordinate setting	Manual guidance coordinate system setting	SC, Guidance coordinates
21	FD	Feed hold	Feed hold	FD
22	SF	Overall system status	Request for high-speed status information	SF
23	MR	Variable read	Global definition variable read	MR, Name, Type
24	IW	I/O write	Compulsive I/O write	IW, Wire No., Status
25	VR	Version read	Robot and software version	VR
26	DO	DO statement run	DO statement run	DO, Statement

Table 2 Command list (Robot controller → Host computer)

No.	Command	Description	Details	Text
1	FL (*1)	File		FL, File content
2	OK	Acknowledge		OK
3	NG	Non-acknowledge		NG

*1: The following files can be transmitted by the FL command.
(1) RAM file [User file (program + position data), parameter file]
(2) File directory
(3) Status
(4) Error history
(5) Overall system status
(6) Version information
(7) Variable read data

*2: The internal command that can be executed by the EC command are as follows:
(1) MODE
(2) OVRD
(3) BREAK

*3: The RN command can be turned ON and OFF by Ethernet parameter setting.